

E-Commerce Sales Analytics — SQL Business Insights

Analysis of the Olist Brazilian E-Commerce dataset (99K+ orders) using MySQL

Query 1: How many total orders, and how many actually got delivered?

```
mysql> SELECT
  ->     order_status,
  ->     COUNT(*) AS total_orders
  -> FROM orders
  -> GROUP BY order_status
  -> ORDER BY total_orders DESC;
```

order_status	total_orders
delivered	96478
shipped	1107
canceled	625
unavailable	609
invoiced	314
processing	301
created	5
approved	2

8 rows in set (0.54 sec)

Insight: 97% of orders (96,478 / 99,441) were successfully delivered. This confirmed filtering to `order_status = 'delivered'` for all revenue calculations, so canceled/unfulfilled orders don't inflate the numbers.

Query 2: Monthly Revenue Trend

```
mysql> SELECT
->     DATE_FORMAT(o.order_purchase_timestamp, '%Y-%m') AS month,
->     ROUND(SUM(p.payment_value), 2) AS revenue
-> FROM orders o
-> JOIN payments p ON o.order_id = p.order_id
-> WHERE o.order_status = 'delivered'
-> GROUP BY month
-> ORDER BY month;
+-----+-----+
| month | revenue |
+-----+-----+
| 2016-10 | 46566.71 |
| 2016-12 | 19.62 |
| 2017-01 | 127545.67 |
| 2017-02 | 271298.65 |
| 2017-03 | 414369.39 |
| 2017-04 | 390952.18 |
| 2017-05 | 567066.73 |
| 2017-06 | 490225.60 |
| 2017-07 | 566403.93 |
| 2017-08 | 646000.61 |
| 2017-09 | 701169.99 |
| 2017-10 | 751140.27 |
| 2017-11 | 1153528.05 |
| 2017-12 | 843199.17 |
| 2018-01 | 1078606.86 |
| 2018-02 | 966510.88 |
| 2018-03 | 1120678.00 |
| 2018-04 | 1132933.95 |
| 2018-05 | 1128836.69 |
| 2018-06 | 1012090.68 |
| 2018-07 | 1027903.86 |
| 2018-08 | 985414.28 |
+-----+-----+
22 rows in set (1.45 sec)

mysql>
```

Insight: Revenue grew roughly 9x from Jan 2017 (R\$127K) to Nov 2017 (R\$1.15M), then plateaued between R\$1.0M–1.13M/month through mid-2018 — suggesting a shift from a growth phase to a maturity phase

Query 3: Top 10 product categories by revenue

```
mysql> SELECT
->   pr.product_category_name,
->   ROUND(SUM(oi.price), 2) AS total_sales,
->   COUNT(DISTINCT oi.order_id) AS num_orders,
->   ROUND(SUM(oi.price) / COUNT(DISTINCT oi.order_id), 2) AS avg_order_value
-> FROM order_items oi
-> JOIN products pr ON oi.product_id = pr.product_id
-> JOIN orders o ON oi.order_id = o.order_id
-> WHERE o.order_status = 'delivered'
-> GROUP BY pr.product_category_name
-> ORDER BY total_sales DESC
-> LIMIT 10;
```

product_category_name	total_sales	num_orders	avg_order_value
beleza_saude	1233131.72	8647	142.61
relogios_presentes	1166176.98	5495	212.23
cama_mesa_banho	1023434.76	9272	110.38
esporte_lazer	954852.55	7530	126.81
informatica_acessorios	888724.61	6530	136.10
moveis_decoracao	711927.69	6307	112.88
utilidades_domesticas	615628.69	5743	107.20
cool_stuff	610204.10	3559	171.45
automotivo	578966.65	3810	151.96
brinquedos	471286.48	3804	123.89

10 rows in set (3.67 sec)

Insight: beleza_saude leads through volume (8,647 orders), while relogios_presentes ranks #2 with nearly 2x the average order value (R\$212 vs. ~R\$110 for high-volume categories) — showing revenue comes from two different strategies: volume-driven vs. value-driven categories.

Query 4: Average review score by category

```
mysql> SELECT
->     pr.product_category_name,
->     ROUND(AVG(r.review_score), 2) AS avg_rating,
->     COUNT(r.review_score) AS num_reviews
-> FROM reviews r
-> JOIN order_items oi ON r.order_id = oi.order_id
-> JOIN products pr ON oi.product_id = pr.product_id
-> GROUP BY pr.product_category_name
-> HAVING num_reviews >= 50
-> ORDER BY avg_rating DESC
-> LIMIT 10;
```

product_category_name	avg_rating	num_reviews
livros_interesse_geral	4.45	549
construcao_ferramentas_ferramentas	4.44	99
livros_importados	4.40	60
livros_tecnicos	4.37	266
alimentos_bebidas	4.32	279
malas_acessorios	4.32	1088
portateis_casa_forno_e_cafe	4.30	76
fashion_calcados	4.23	261
alimentos	4.22	495
cine_foto	4.21	73

10 rows in set (2.64 sec)

Insight: Book categories dominate satisfaction (3 of top 4 spots) despite not appearing in the top-10 revenue list — a satisfaction-revenue mismatch that could point to a loyalty/cross-sell opportunity for a smaller but highly-satisfied segment.

Query 5: Delivery delay analysis (actual vs. estimated delivery)

```
mysql> SELECT
->     CASE
->     WHEN order_delivered_customer_date > order_estimated_delivery_date THEN 'Late'
->     ELSE 'On Time or Early'
->     END AS delivery_status,
->     COUNT(*) AS num_orders,
->     ROUND(AVG(DATEDIFF(order_delivered_customer_date, order_estimated_delivery_date)), 1) AS avg_days_diff
-> FROM orders
-> WHERE order_status = 'delivered'
->     AND order_delivered_customer_date IS NOT NULL
-> GROUP BY delivery_status;
```

delivery_status	num_orders	avg_days_diff
On Time or Early	88644	-13.7
Late	7826	8.9

2 rows in set (0.73 sec)

Insight: 92% of deliveries arrived on time or early, averaging nearly two weeks ahead of the estimate. The remaining 8% arrived ~9 days late on average — suggesting delivery estimates are set conservatively to buffer against that slower tail.